

How Many In-Context Examples Is a Task Description Worth?

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Abstract

Prompts combine a task description with in-context examples, but theories of in-context learning (ICL) model only the examples. Under the Bayesian account of ICL, the examples are observations and the description is the prior, and Bayesian statistics measures a prior by its effective sample size (ESS): the number of observations it is equivalent to. We define the ESS of a task description as the number of in-context examples at which a Bayes-optimal predictor attains the same one-step log-loss regret as it attains from the description alone, and compute it in in-context linear regression, where the Bayes-optimal predictor is exact, as a function of the description’s precision and reliability. Two results. A reliable description that removes a fraction $1 - r$ of the prior variance is worth about $d(1 - r)$ examples in a d -dimensional task. An unreliable description, correct with probability p , has a regret floor of $(1 - p)$ times the no-description regret plus the entropy of p , so its ESS saturates in precision: at $p = 0.9$, a fiftyfold increase in precision adds a few examples where a reliable description gains a hundred. [Third, whether meta-trained Transformers reproduce these numbers.] Our claims concern Bayes-optimal predictors and small meta-trained sequence models, not large language models.

1 Introduction

Theoretical work on in-context learning has modelled prompts as input–output examples alone (Garg et al., 2022; Akyürek et al., 2023; Xie et al., 2022; Genewein et al., 2025). Prompts in practice also carry a natural-language task description (Brown et al., 2020), and the empirical record on its value is inconsistent: a description can substitute for hundreds of labelled examples (Le Scao and Rush, 2021; Honda et al., 2025), yet models also discount instructions in favour of examples (Gupta et al., 2025; Webson and Pavlick, 2022). Recent theory admits descriptions into the model (Huang and Ge, 2025; Lin et al., 2025; Tong et al., 2026) but does not express their value on the scale of the examples they accompany.

We propose that scale. Under the Bayesian account of ICL (Xie et al., 2022), a sequence model approximates the posterior predictive for a latent task: the in-context examples are observations and the task description specifies the prior. Bayesian statistics measures a prior on the scale of observations by its *prior effective sample size* (Morita et al., 2008), and two properties of a prior govern that measure: how concentrated it is, and whether it is compatible with the data (Evans and Moshonov, 2006). A task description has the same two properties, which we call its **precision**, the posterior contraction it induces, and its **reliability**, the prior probability that it is correct, modelled with a robust mixture prior (Schmidli et al., 2014).

We compute the ESS of a description as a function of both in in-context linear regression (Garg et al., 2022), where the Bayes-optimal predictor is available in closed form, for a single prediction, the ordinary use of a prompt.

Contributions.

1. A definition of the ESS of a task description by matching one-step log-loss regret against in-context examples (§3). A prediction-based ESS was posed as an open problem by Reimherr et al. (2021); in the linear-Gaussian setting ours is exact.
2. For reliable descriptions, $\text{ESS} \approx d(1 - r)$ (§4).
3. For unreliable descriptions, the regret has a floor of $(1 - p)$ times the no-description regret plus $H(p)$, so the ESS saturates in precision (§5).
4. [Pending.] Whether meta-trained Transformers reproduce these ESS values (§6).

2 Related work

Bayesian ICL. Xie et al. (2022) derive ICL as implicit Bayesian inference over a latent concept. In linear regression, Transformers meta-trained on Gaussian tasks track the Bayes-optimal ridge predictor (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024), and Genewein et al. (2025) evaluate small meta-trained networks by log-loss regret against the Bayes-optimal predictor, the criterion we adopt. None of these expresses a prior on the scale of in-context examples.

Task descriptions in ICL theory. Huang and Ge (2025) add a descriptor token to in-context linear regression, the mean of the input distribution, and show that linear self-attention uses it to centre the inputs. That descriptor is independent of the regression weights, so a Bayes-optimal predictor assigns it zero ESS; ours specifies the prior on the weights. Lin et al. (2025) supply a hypothesis class as a prefix; Tong et al. (2026) show that an incorrect instruction is overridden at an exponential rate in the number of demonstrations, without a reliability parameter; Lin and Lee (2024) analyse a Gaussian-mixture task prior, whose posterior is the same conjugate computation as ours.

Prior effective sample size. Morita et al. (2008) define the ESS of a prior by curvature matching and Reimherr et al. (2021) by prior–likelihood discordance, naming a prediction-based measure as open. Robust mixture priors (Schmidli et al., 2014) arise in clinical trials with historical controls, where the informative component may conflict with the data.

3 Setting

Task distribution. A task is a weight vector $w \in \mathbb{R}^d$. An in-context example is an input $x \sim \mathcal{N}(0, I_d)$ with label $y = w^\top x + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma_y^2)$. The base prior is $w \sim \mathcal{N}(0, s_0^2 I_d)$. We set $\sigma_y = 1$ and write $a_0 = s_0^2/\sigma_y^2$ for the prior signal-to-noise ratio.

Task description. A description is a vector $m \in \mathbb{R}^d$ with a variance $s_\ell^2 = r s_0^2$, $r \in (0, 1)$, the *precision ratio*; conditional on the description being correct, $w \sim \mathcal{N}(m, s_\ell^2 I_d)$. It is generated hierarchically, $m \sim \mathcal{N}(0, (1-r)s_0^2 I_d)$, so that the marginal law of w is the base prior whether or not a description is present. The description is correct with probability p , its *reliability*; otherwise $w \sim \mathcal{N}(0, s_0^2 I_d)$ independently of m . The Bayes-optimal prior given a description is the robust mixture $p\mathcal{N}(m, s_\ell^2 I) + (1-p)\mathcal{N}(0, s_0^2 I)$ of Schmidli et al. (2014). The description idealises an instruction as a vector: the learner must weight it, not parse it.

Regret. The predictor outputs a predictive distribution for the label of one query x . Its regret is its expected log loss minus that of the oracle that knows w , in nats (Genewein et al., 2025), the expectation over tasks, inputs, and labels.

Definition 1 (Effective sample size of a description). *Let R^{desc} be the regret of the Bayes-optimal predictor conditioned on the description and no examples, and $R^{\text{ex}}(n)$ that of the Bayes-optimal predictor under the base prior conditioned on n examples. The ESS of the description is the n , interpolated linearly between integers, at which $R^{\text{ex}}(n) = R^{\text{desc}}$.*

Both predictors are the same Bayes-optimal predictor with different conditioning information, so the ESS is a statement about information rather than about a learner.

4 Precision

For a reliable description ($p = 1$) the posterior covariance of w after inputs $X_n \in \mathbb{R}^{n \times d}$ under a Gaussian prior with signal-to-noise ratio a is $\Sigma = \sigma_y^2(I/a + X_n^\top X_n)^{-1}$, independent of m and of the labels, and the one-step regret is $\frac{1}{2}\mathbb{E} \log(1 + x^\top \Sigma x)$. The description alone has $a = r a_0$ and $n = 0$; the examples have $a = a_0$.

Proposition 1. *At high signal-to-noise ratio and for $\text{ESS} < d$, $\text{ESS} \approx d(1-r)$.*

Sketch. With $n < d$ examples at high signal-to-noise ratio, the posterior variance is near zero along the n observed directions and s_0^2 along the other $d - n$, so $x^\top \Sigma x \approx s_0^2 \|x_\perp\|^2$ with $\|x_\perp\|^2 \sim \chi_{d-n}^2$ and the regret is about $\frac{1}{2} \log(a_0(d-n))$. The description contracts every direction to $r s_0^2$, giving about $\frac{1}{2} \log(r a_0 d)$. Equating, $d - n = r d$.
[Write out with the error term.]

Table 1: ESS of a reliable description, $a_0 = 10$, three seeds; standard deviation across seeds below 2% of each value. The approximation $d(1 - r)$ holds while the ESS is below d .

d	r	ESS	$d(1 - r)$
4	0.5	1.77	2.0
4	0.2	3.32	3.2
16	0.5	7.62	8.0
16	0.2	12.6	12.8
16	0.05	16.9	15.2
16	0.01	26.1	15.8
64	0.5	31.6	32.0
64	0.2	50.7	51.2
64	0.05	62.1	60.8
64	0.01	73.5	63.4

Table 2: ESS of an unreliable description, $d = 16$, $a_0 = 10$, calibrated predictor; standard deviation across three seeds at most 1% of each value.

r	$p = 1$	$p = 0.99$	$p = 0.9$	$p = 0.7$
0.5	7.6	7.5	6.3	4.1
0.05	16.9	16.6	14.8	11.7
0.001	116.5	64.2	23.1	15.6

Table 1 compares the approximation with the exact ESS. A description that removes half the prior variance is worth half the dimension in examples. Once the description is precise enough to be worth more than d examples, the approximation stops applying and the exact ESS grows beyond d , because d noisy examples do not determine w exactly.

5 Reliability

Given an unreliable description and no examples, the Bayes-optimal predictive distribution is the p -mixture of the two components’ predictives, because nothing has been observed that could identify the correct component.

Proposition 2. *As $r \rightarrow 0$, the regret of a description with reliability p tends to $(1 - p) R^{\text{ex}}(0) + H(p)$, where $H(p)$ is the binary entropy and $R^{\text{ex}}(0)$ the regret with neither description nor examples.*

Sketch. As $r \rightarrow 0$ the two components’ predictives separate: under a correct description the mixture assigns the label density p times the oracle’s, costing $\log(1/p)$; under an incorrect one it assigns $(1 - p)$ times the base predictive’s, costing $R^{\text{ex}}(0) + \log(1/(1 - p))$. Weighting by p and $1 - p$ gives the claim. *[Write out; the limit is approached from above.]* At $d = 16$, $a_0 = 10$, $R^{\text{ex}}(0) = 2.51$ nats, and the formula gives 0.58 at $p = 0.9$ against a computed 0.57 at $r = 0.001$.

Since $R^{\text{ex}}(n)$ is decreasing in n , the ESS of an unreliable description is bounded by the n at which $R^{\text{ex}}(n)$ reaches this floor, however precise the description.

Table 2 shows the saturation. A fiftyfold increase in precision (r from 0.05 to 0.001) raises the ESS from 16.9 to 116.5 for a reliable description, from 14.8 to 23.1 at $p = 0.9$, and from 11.7 to 15.6 at $p = 0.7$. A reliability of 0.99 already halves the ESS of the most precise description. The same pattern holds at $d \in \{4, 64\}$ and $a_0 = 100$.

6 Meta-trained Transformers

[Results pending. The design below is fixed except where marked.]

Design. We meta-train Transformers on prompts sampled from the generative process of §3 and evaluate them against the Bayes-optimal predictor on held-out prompts, following Genewein et al. (2025). The prompt uses the prefix embedding of Huang and Ge (2025): an optional descriptor token carrying m , up to 15 example tokens, and a query

token, with two indicator coordinates distinguishing the token types. The model has no positional encoding, so the examples are exchangeable, as for the Bayes-optimal predictor. Architecture and optimisation follow Garg et al. (2022): 12 layers, 8 heads, embedding width 256, Adam at 10^{-4} , fresh prompts at every step. *[Output and loss (a predictive distribution trained by log loss, as in Genewein et al. (2025), or a point prediction trained by squared error, as in Garg et al. (2022)) and the scoring of prompt positions to be fixed.]* The descriptor token carries only m ; r and p are fixed per model and never stated, so the network infers both from the meta-training distribution, as a language model would from pretraining. We use $d = 5$, $a_0 = 10$, and half of the training prompts carry no descriptor.

Readouts and order. For each model we report the regret after n examples with and without a descriptor, and the descriptor’s ESS. The experiments vary one factor at a time: a reliable precise descriptor ($r = 0.05$); the same model evaluated on descriptors with $p = 0.9$; a coarse descriptor ($r = 0.5$); three seeds; and only then meta-training at $p = 0.9$.

Bayes-optimal targets. At this setting the ESS is 6.7 for the precise descriptor and 2.2 for the coarse one under log loss (7.4 and 2.7 under squared error); at $p = 0.9$ these become 5.0 and 1.8 (4.4 and 2.1).

[Table: network versus Bayes-optimal ESS and regret curves. Figure: regret against n .]

7 Discussion

A task description is worth a definite number of in-context examples to a Bayes-optimal predictor, and that number has a simple form: the fraction of the prior variance the description removes, times the dimension of the task, provided the description can be trusted. Reliability caps what precision can buy, because a predictor that cannot yet tell whether the description is correct must hedge between it and the base prior, and the cost of hedging does not shrink as the description sharpens.

Limitations. Our claims concern Bayes-optimal predictors and small meta-trained sequence models, not language models. The description is a vector rather than text, and the setting is linear regression with Gaussian inputs, chosen because the Bayes-optimal predictor is exact. The ESS is defined for a single prediction; with sequential feedback the predictor can identify whether a description is correct, and the ESS changes with the prediction horizon, which we leave to future work. *[Network results pending.]*

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